Sanjeevani

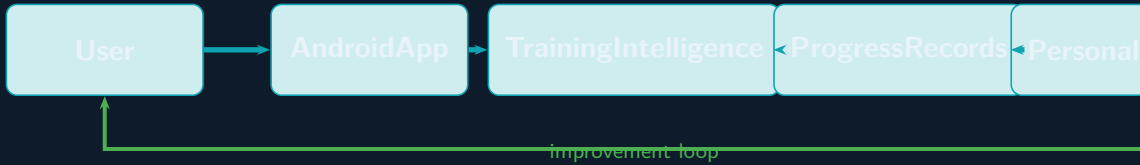
How the System Works

From user action to feedback, progress, and personalized practice

Stack:
Android (Kotlin / Compose)
MediaPipe · CameraX
Android TTS · STT
SharedPreferences
Fully on-device

The simplest possible explanation

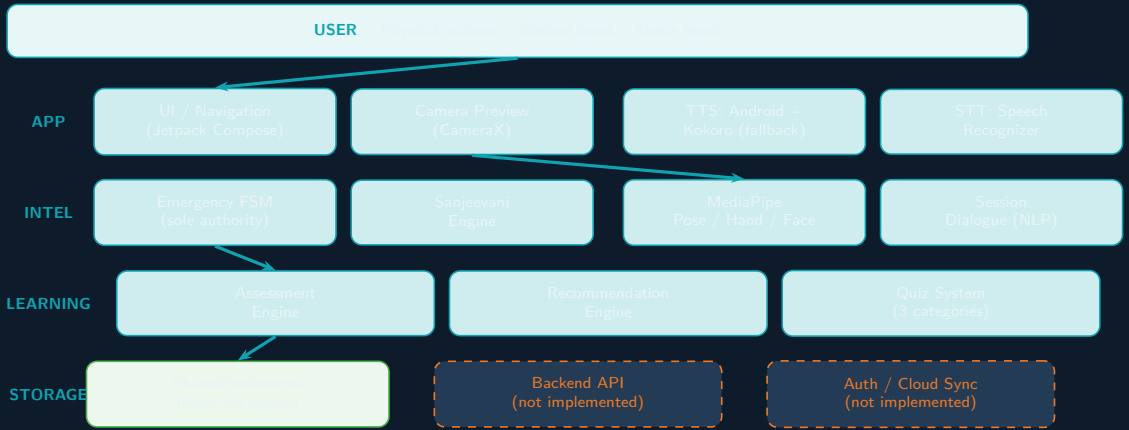
Five blocks, one sentence



"The user practises a skill. The app evaluates supported actions. Feedback helps the user improve. Recorded progress guides the next session."

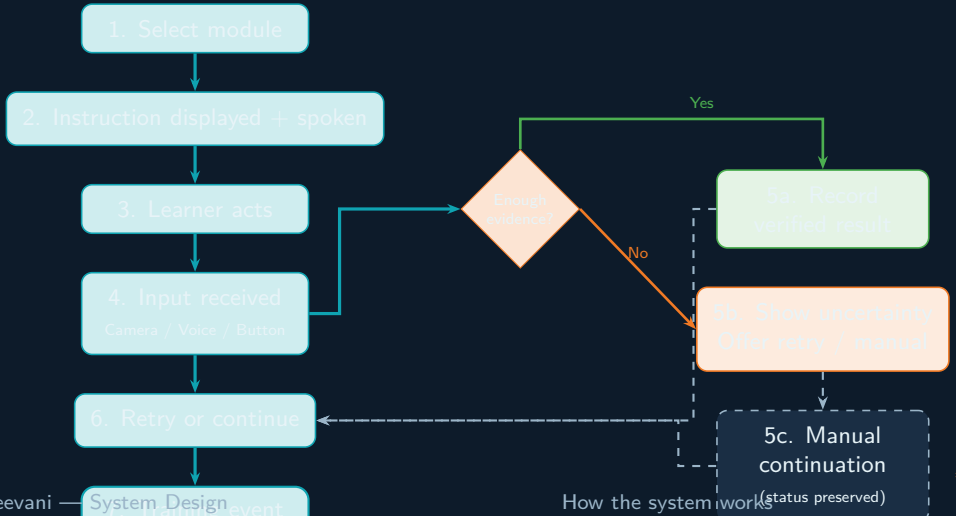
High-level architecture

Five layers — solid = implemented dashed = proposed/not built



What happens during one training session?

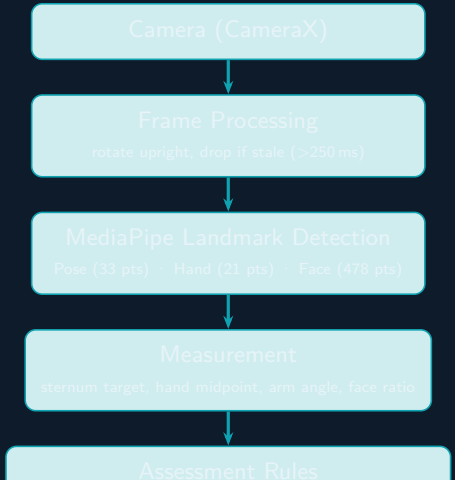
Step-by-step flow from module selection to recorded outcome



Manual continue ≠ verified success

How camera feedback works

From pixels to supported corrective guidance



Four possible assessment states:

✓ **Verified:** confidence >65%, stable ≥1 s

~ **Detected but incorrect:** target found, wrong position

? **Inconclusive:** low confidence or tracking lost

× **Unsupported:** depth, force, recoil, contact

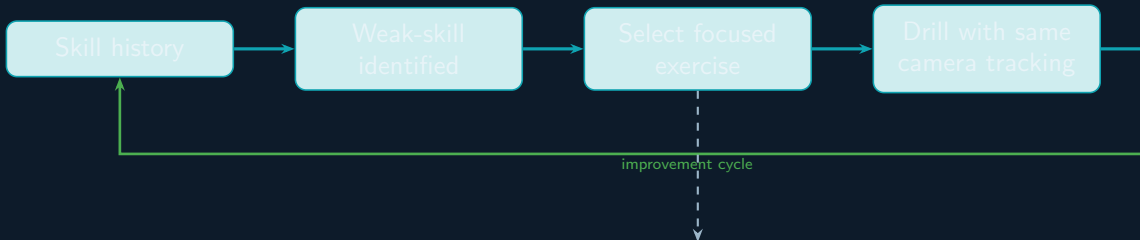
module;
[dashedbox,minimum width=2.0cm,minimum height=0.9cm,font=] (aed) at (-6.5,-4.8) AED (not impl.); [dashedbox,minimum width=2.0cm,minimum height=0.9cm,font=] (cho) at (-3.8,-4.8) Choking (not impl.); [dashedbox,minimum width=2.0cm,minimum height=0.9cm,font=] (rec) at (-1.1,-4.8) Recovery Pos. (not impl.);

[text=TextSub,font=] at (3.0,-4.8) UI cards present; available=false; not dispatched.;

(labelled as demo);

How targeted practice works

From skill history to the right exercise — reusing the same mechanism



CPR hand-placement:

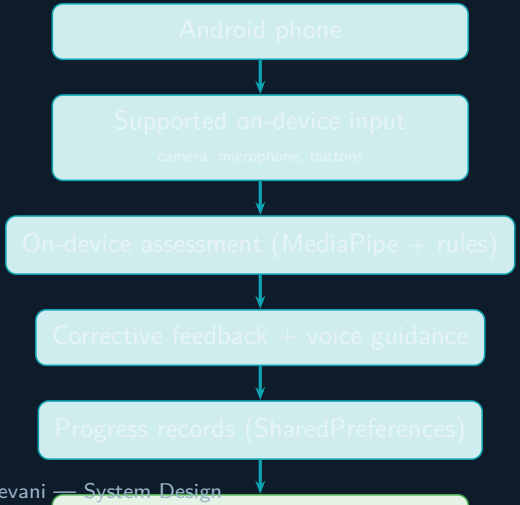
- criterion: `hand_placement`
- drill: skip engine to HAND_POSITIONING state
- same SpatialReasoner + AR arrow + voice cue
- no new code required

Allergy EpiPen:

- criterion: `epipen`
- drill: allergy_seq route
- 6-step panel + co
- AllergyPositioning wrist-to-thigh gu

How Sanjeevani can reduce equipment dependence

What a phone can support — and what it cannot replace



- ✓ Repeat practice sessions without scheduling
- ✓ Skill-targeted revision between workshops
- ✓ No internet or account required
- ✓ Progress persists across sessions on device

What it does not replace

Digital feedback is limited to supported, validated observations. Physical mannequins and qualified instructors remain important wherever hands-on verification or supervised training is required.

Data, privacy, and reliability

How data flows, what is stored, and how edge cases are handled

Camera frames	On-device only; not stored or transr
Voice input	Android STT; final transcript only
TTS output	Android TTS + Kokoro; local audic
Training events	SharedPreferences (local)
Quiz attempts	SharedPreferences (local)
Network calls	None — fully offline
Raw media stored	No

`CAMERA` — CameraX preview + MediaPipe
`RECORD_AUDIO` — Speech recognizer
No location, contacts, or storage permissions
Permissions requested at runtime with rationale

Situation	Behaviour
Tracking lost	Rate/cycle invalidated; target frozen
Multi-person frame	Treated as ambiguous; no assessment
Stale camera frame	Dropped (>250 ms age gate)
Stale STT result	TurnGate (epoch + version) blocks it
App paused	Active turn cancelled; retry available
Process killed	In-memory session lost; store data kept
Recognition error	Retry prompt + tap-button fallback
Duplicate answer	Version guard prevents

1. Training is interactive

2. Feedback is based on available evidence

3. Progress helps identify what to practise next

Technical references I

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